

Portfolio - Prompt Evaluation Studio Pro

Portfolio Content — Prompt Evaluation Studio Pro

About Me

Passionate AI/ML developer with hands-on experience in Python, prompt engineering, and building production-quality AI applications. I enjoy solving real problems at the intersection of software engineering and artificial intelligence.

Skills Demonstrated

- Python 3.12, Type Hints, Pydantic
- Streamlit UI Development
- Google Gemini API / google-genai SDK
- Prompt Engineering (Role, CoT, JSON, Chaining)
- SQLAlchemy, SQLite Database Design
- Docker, pytest, CI-ready Testing
- PDF/CSV/JSON Export, Plotly Analytics
- Modular Architecture, Error Handling

Problem Statement

Prompt engineering is critical for AI applications, yet most developers lack tools to systematically create, test, compare, evaluate, and version-control prompts. Ad-hoc testing in chat interfaces doesn't scale and doesn't provide metrics.

Business Problem

Organizations deploying LLMs need:

- Consistent prompt quality assurance
- Version control for prompt changes
- Comparative testing before production
- Automated evaluation metrics
- Audit trails of prompt experiments

Without these, teams face inconsistent AI outputs, untracked prompt changes, and no measurable quality standards.

Solution

Prompt Evaluation Studio Pro provides an end-to-end platform where teams can:

1. Create and configure prompts with full parameter control
2. Compare multiple prompt variants objectively
3. Automatically evaluate outputs on 9 criteria

4. Optimize prompts with AI-generated improvements
5. Track experiments and analytics over time
6. Export reports for stakeholders

Architecture

Layered architecture: Streamlit Frontend → Service Factory → Services (Auth, Prompts, Gemini, Evaluation, etc.) → SQLite Database + Gemini API.

See ARCHITECTURE.md and assets/diagrams/ for detailed diagrams.

Workflow

Sign Up → Configure API → Create/Load Prompts → Run in Playground → Compare Variants → Evaluate Quality → Optimize → Version Control → Export Reports → Analyze Trends

Features (12 Modules)

1. Authentication, 2. Dashboard, 3. Playground, 4. Comparison, 5. Version Control, 6. Library, 7. Experiments, 8. Evaluation Engine, 9. Optimizer, 10. Analytics, 11. Export, 12. Admin Settings

Technology Stack

Python 3.12 | Streamlit | Google Gemini | SQLite | SQLAlchemy | Pydantic | Pandas | Plotly | Docker | pytest

Database Design

7 tables with proper relationships: Users → Prompts → Versions, Experiments, Evaluations, Analytics.
See DATABASE.md.

Prompt Engineering Techniques

Role Prompting, Structured JSON, Chain of Thought, Few-Shot Templates, Prompt Chaining, Output Validation, Error Recovery, Weighted Scoring, Variable Templates, Constraint Specification

Challenges

1. ****Consistent Evaluation Scoring**** — Solved with structured JSON prompts and weighted averaging
2. ****Streaming in Streamlit**** — Solved with placeholder updates and stop flag
3. ****Version Diff Display**** — Built custom line-level diff utility
4. ****Testable Gemini Integration**** — Mock-based pytest without live API

Learnings

- Prompt engineering is an engineering discipline requiring tooling, metrics, and process
- Service layer architecture makes AI apps testable and maintainable
- Structured JSON output is essential for reliable AI pipelines
- User experience matters even for developer tools

My Contribution

Designed and implemented the entire project solo:

- Architecture and database schema
- All 12 feature modules
- Gemini integration with error handling
- Evaluation and optimization prompt design
- Test suite and documentation
- Docker deployment configuration

Results

- 23/23 tests passing
- 12 fully functional modules
- 7-table normalized database
- 9-criteria evaluation engine
- 5 export formats
- Complete documentation suite

Future Improvements

- Multi-LLM support (OpenAI, Claude, Llama)
- Team collaboration and shared workspaces
- Automated prompt regression testing
- CI/CD integration for prompt pipelines
- PostgreSQL for production scale

Impact

Demonstrates ability to build real AI products beyond simple chatbots — showing prompt engineering expertise that employers value for AI/ML roles.

GitHub Description

Professional Prompt Engineering platform built with Python, Streamlit, and Google Gemini. Create, compare, evaluate, optimize, and manage AI prompts with version control, analytics, and automated scoring.

LinkedIn Project Description

Built Prompt Evaluation Studio Pro — a full-stack Prompt Engineering platform using Python, Streamlit, and Google Gemini API. Features include prompt playground with streaming, multi-prompt comparison,

AI-powered evaluation (9 criteria), automatic optimization, version control, experiment tracking, analytics dashboard, and multi-format export. Demonstrates modular architecture, SQLAlchemy database design, Pydantic validation, Docker deployment, and comprehensive pytest coverage.

Resume Project Description

****Prompt Evaluation Studio Pro**** | Python, Streamlit, Gemini API, SQLite, Docker

- Developed a 12-module prompt engineering platform with authentication, version control, and analytics
- Implemented AI evaluation engine scoring outputs on 9 criteria using structured JSON prompts
- Built prompt comparison system supporting 2-5 variants with weighted winner selection
- Designed modular service architecture with 23 passing pytest tests and Docker deployment

STAR Method Explanation

****Situation:**** Needed a portfolio project demonstrating prompt engineering beyond basic API integration.

****Task:**** Build a production-quality platform covering the full prompt lifecycle with evaluation metrics.

****Action:**** Designed layered architecture, implemented 12 modules, created structured evaluation prompts, wrote comprehensive tests and documentation, containerized with Docker.

****Result:**** Fully functional platform with 23 passing tests, 9-criteria evaluation engine, and complete documentation — ready to demonstrate in interviews.